Wiring Guide — Smart Security System

Target hardware: Arduino Uno R3

Project: Smart Security System (HC-SR04 + ST7735S TFT + Buzzer + LEDs + Button)

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1. Full Pin-Assignment Table

Signal Connections

Arduino Uno Pin	Component	Pin / Label	Direction	Notes
8	TFT LCD	RES	OUTPUT	Hardware reset, active LOW
9	TFT LCD	DC (A0/RS)	OUTPUT	Data/Command select
10	TFT LCD	CS (TCS)	OUTPUT	Chip-select, active LOW

| **5** | HC-SR04 | TRIG | OUTPUT | 10 µs HIGH trigger pulse |

| **6** | HC-SR04 | ECHO | INPUT | Echo pulse, measured by pulseIn |

| **3** | Red LED | Anode (+) | OUTPUT | 330 Ω series resistor to GND |

| **4** | Green LED | Anode (+) | OUTPUT | 330 Ω series resistor to GND |

| **7** | Active Buzzer | + (signal) | OUTPUT | Active type (oscillator built-in) |

| **2** | Push Button | Leg A | INPUT | INPUT_PULLUP; press reads LOW |

| **11** | TFT LCD | SDA | OUTPUT | Hardware SPI MOSI |

| **13** | TFT LCD | SCL | OUTPUT | Hardware SPI SCK |

Power and Ground Connections

Arduino Uno Rail	Component	Component Pin	Notes
5 V	TFT LCD	VCC	Power
5 V	TFT LCD	BL	Backlight — always on
GND	TFT LCD	GND	
5 V	HC-SR04	VCC	
GND	HC-SR04	GND	
GND	Red LED	Cathode (−)	Via 330 Ω resistor to GND
GND	Green LED	Cathode (−)	Via 330 Ω resistor to GND
GND	Active Buzzer	− (GND)	
GND	Push Button	Leg B	

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2. Wire-Colour Suggestion Table

Using consistent wire colours makes troubleshooting much easier. The suggestions below follow a common convention for this type of project.

Arduino Uno Pin	Connects To	Suggested Wire Colour
8	TFT RES	White
9	TFT DC	Yellow
10	TFT CS	Orange
5	HC-SR04 TRIG	Blue
6	HC-SR04 ECHO	Purple
3	Red LED anode (+)	Red
4	Green LED anode (+)	Green
7	Buzzer + (signal)	Black
2	Button leg A	Grey
11	TFT SDA	Blue
13	TFT SCL	White

> ***Tip:** The TFT SPI data lines (pins 11 and 13) share the same suggested colours as some other connections. Consider using a slightly different shade or labelling the ends of your wires to avoid confusion during assembly.*

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3. LED Resistor Calculation

Both LEDs require a current-limiting resistor in series between their cathode and GND. Using the wrong resistor (or omitting it) can damage the LED or the Arduino output pin.

Formula

```

R = (V_supply − V_f) / I_f

```

Parameter	Value
V_supply	5 V (Arduino Uno digital output)
V_f (Red)	≈ 2.0 V
V_f (Green)	≈ 2.2 V
I_f target	15 mA (0.015 A)

Calculation (worst case — Red LED, lowest V_f)

```

R = (5 V − 2.1 V) / 0.015 A

R = 2.9 V / 0.015 A

R ≈ 193 Ω (for 220 Ω)

```

Use 330 Ω — the nearest standard resistor value above 193 Ω (for 220 Ω).

A 330 Ω resistor is safe for both the red and green LED at this supply voltage. It keeps the current comfortably below the maximum rated current for typical 5 mm LEDs and well within the Arduino Uno's 40 mA per-pin limit.

Wiring the resistors

Place each 330 Ω resistor between the LED's **cathode (shorter leg / flat side)** and the GND rail on your breadboard. The LED's anode (longer leg) connects directly to the Arduino signal pin.

```

Arduino Pin 3 —▶ Red LED (+) —▶ Red LED (−) —▶ [330 Ω] —▶ GND

Arduino Pin 4 —▶ Green LED (+) —▶ Green LED (−) —▶ [330 Ω] —▶ GND
```

4. Button Pull-Up Wiring

The push button uses the Arduino Uno's internal pull-up resistor, so no external resistor is needed.

How to wire it

```

Arduino Pin 2 —— Button Leg A

GND          —— Button Leg B
```

- Connect one leg of the button to **pin 2**.
- Connect the other leg to **GND**.
- That's it — no resistor, no other connections required.

How it behaves in firmware

The firmware configures pin 2 as `INPUT_PULLUP`. This means:

Button State	Pin 2 Reads
Released	HIGH (≈ 5 V)
Pressed	LOW (≈ 0 V)

The pull-up holds the pin HIGH when the button is open. Pressing the button connects the pin directly to GND, pulling it LOW. The firmware detects this HIGH → LOW transition as a valid press event (after debouncing).

> ***No external pull-up or pull-down resistor is needed** when using `INPUT_PULLUP` mode.*

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5. Fasizi 1.8" TFT Module Notes

Built-in voltage regulation and level shifting

The Fasizi 1.8-inch ST7735S module typically includes:

- A **built-in 3.3 V LDO regulator**, so the VCC pin can accept 3.3 V or 5 V input.
- **Level shifters** on the SPI and control lines, making the module's logic inputs compatible with the Arduino Uno's 5 V signals.

This means it is generally safe to connect the TFT VCC pin to the Uno's **5 V header** rather than 3.3 V. Always confirm this with your specific module's datasheet before powering it at 5 V, as not all ST7735S breakout boards include level shifters.

SPI bus speed

Pins 11 (MOSI) and 13 (SCK) are the Arduino Uno's dedicated **hardware SPI** pins. The `Adafruit_ST7735` library uses hardware SPI for maximum throughput, which is important for fast screen refreshes (e.g. flashing the display during the ALARM state). Do **not** substitute these with arbitrary digital pins; you must use pins 11 and 13 for SPI data.

SPI pin summary for this module

TFT Label	Arduino Uno Pin	Function
SDA	11	Hardware SPI MOSI
SCL	13	Hardware SPI SCK
CS	10	Chip-select (GPIO)
DC	9	Data/Command (GPIO)
RES	8	Reset (GPIO)
VCC	5V	Power
BL	5V	Backlight (always on)
GND	GND	Ground

6. Quick Assembly Checklist

Use this list to verify your wiring before powering up.

- [ ] TFT RES → Pin 8 (white wire)
- [ ] TFT DC → Pin 9 (yellow wire)
- [ ] TFT CS → Pin 10 (orange wire)

- [ ] TFT SDA → Pin 11 (blue wire)
- [ ] TFT SCL → Pin 13 (white wire)
- [ ] TFT VCC → 5 V
- [ ] TFT GND → GND
- [ ] TFT BL → 5 V
- [ ] HC-SR04 TRIG → Pin 5 (blue wire)
- [ ] HC-SR04 ECHO → Pin 6 (purple wire)
- [ ] HC-SR04 VCC → 5 V
- [ ] HC-SR04 GND → GND
- [ ] Red LED anode → Pin 3 (red wire)
- [ ] Red LED cathode → 330 Ω → GND
- [ ] Green LED anode → Pin 4 (green wire)
- [ ] Green LED cathode → 330 Ω → GND
- [ ] Buzzer + → Pin 7 (black wire)
- [ ] Buzzer − → GND
- [ ] Button leg A → Pin 2 (grey wire)
- [ ] Button leg B → GND
- [ ] No external resistor on button (uses INPUT_PULLUP)

7. Using a Prototype Expansion Shield

If you have a **Prototype Expansion Shield** (with I/O ports, 26-pin connector, prototyping area, reset button, and I2C port) sitting on top of your Arduino Uno R3, the wiring is the same — the shield simply passes all Uno pins through to labelled

headers on top. Pin 5 on the Uno is still pin 5 on the shield.

What each shield feature does

Shield Feature	What It Is	Used in This Project?
I/O port headers	All digital (D0-D13) and analog (A0-A5) pins broken out	Yes — all signal pins
26p connector	Passes through all Uno header pins in one block	Yes — passthrough
Prototyping area	Small perfboard area for soldering custom circuits	Optional
Reset button	Resets the Arduino (same as the Uno onboard reset)	Convenient
I2C port	Breaks out SDA (A4) and SCL (A5) for I2C devices	Not used

How to wire each component through the shield

Connect jumper wires from the shield headers to your components on the breadboard.

Match the shield header label to the pin number from the wiring guide:

TFT Display (5 signal + 3 power wires)

TFT Pin	Shield Header	Wire Colour
CS	D10	Orange
DC	D9	Yellow
RES	D8	White

| SDA | D11 | Blue |

| SCL | D13 | White |

| VCC | 5V | Red |

| BL | 5V | Red |

| GND | GND | Black |

HC-SR04 Sensor (4 wires)

| HC-SR04 Pin | Shield Header | Wire Colour |

|---|---|---|

| TRIG | D5 | Blue |

| ECHO | D6 | Purple |

| VCC | 5V | Red |

| GND | GND | Black |

Red LED (with 330 ohm resistor)

```
Shield D3 header —▶ Red LED (+) —▶ Red LED (−) —▶ [330 Ω] —▶ Shield GND
```

Green LED (with 330 ohm resistor)

```
Shield D4 header —▶ Green LED (+) —▶ Green LED (−) —▶ [330 Ω] —▶ Shield GND
```

Active Buzzer (2 wires)

| Buzzer Pin | Shield Header | Wire Colour |

|---|---|---|

| + | D7 | Black |

| – | GND | Black |

Push Button (2 wires, no resistor)

| Button Leg | Shield Header | Wire Colour |

|---|---|---|

| Leg A | D2 | Grey |

| Leg B | GND | Black |

Shield wiring tips

- The **26p connector** is just a passthrough — you don't need to do anything special with it.
- The **I2C port** (SDA/SCL) is not used by this project. Leave it unconnected.
- The **prototyping area** can be used to solder the 330 ohm resistors directly if you prefer a permanent build, but a breadboard works fine for prototyping.
- The **reset button** on the shield works the same as pressing the Uno onboard reset.
- If your shield has any **built-in components** (LEDs, buttons, buzzer, display) already soldered to the board, check whether they share pins with this project. Built-in components on D2, D3, D4, D5, D6, D7, D8, D10, D11, or D13 would conflict.

Wiring guide for Smart Security System — Arduino Uno R3

Satisfies Requirements 11.1, 11.2, 11.3, 11.4, 11.5, 11.6